



Review

State-of-the-art of the pyrolysis and co-pyrolysis of food waste: Progress and challenges



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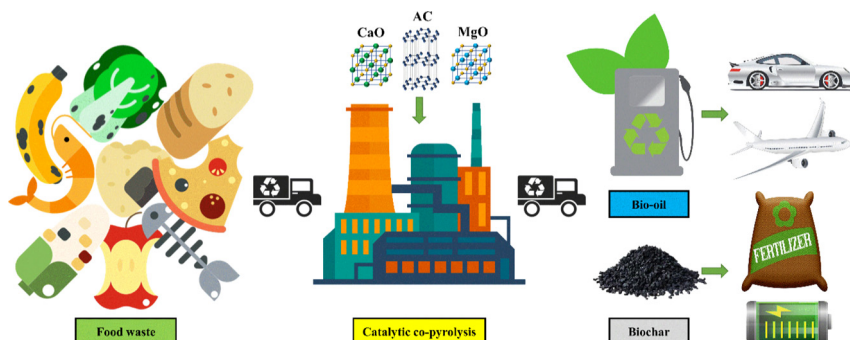
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HIGHLIGHTS

- Pyrolysis is an effective method to convert food waste into valuable products.
- Tire waste is currently the optimal co-feedstock for the pyrolysis of food waste.
- CaO exhibits the best catalytic effect in the pyrolysis of food waste.
- Small distributed plants are more economical and feasible in developing countries.

GRAPHICAL ABSTRACT



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ABSTRACT

The continuous growth of population and the steady improvement of people's living standards have accelerated the generation of massive food waste. Untreated food waste has great potential to harm the environment and human health due to bad odor release, bacterial leaching, and virus transmission. However, the application of traditional disposal techniques like composting, landfilling, animal feeding, and anaerobic digestion are difficult to ease the environmental burdens because of problems such as large land occupation, virus transmission, hazardous gas emissions, and poor efficiency. Pyrolysis is a practical and promising route to reduce the environmental burden by converting food waste into bioenergy. This paper aims to analyze the characteristics of food waste, introduce the production of biofuels from conventional and advanced pyrolysis of food waste, and provide a basis for scientific disposal and sustainable management of food waste. The review shows that co-pyrolysis and catalytic pyrolysis significantly impact the pyrolysis process and product characteristics. The addition of tire waste promotes the synthesis of hydrocarbons and inhibits the formation of oxygenated compounds efficiently. The application of calcium oxide (CaO) exhibits good performance in the increment of bio-oil yield and hydrocarbon content. Based on this literature review, pyrolysis can be considered as the optimal technique for dealing with food waste and producing valuable products.

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Abbreviations: AC, activated carbon; DB, dry basis; EHI, effective hydrogen index; FC, fixed carbon; HHV, higher heating value; LHV, lower heating value; MO, moisture; PAHs, polycyclic aromatic hydrocarbons; PVC, polyvinyl chloride; TGA, thermogravimetric analyzer; VM, volatile matter.

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1. Introduction

Food waste, also known as kitchen waste, is a kind of biodegradable waste that mainly comes from various sources, such as households, food processing, and catering industries (Brennan et al., 2021; Gustafsson et al., 2011). Food waste occurs at every stage of the supply chain, including production, harvesting, storage, transportation, and cooking. The Food and Agriculture Organization has reported that more than 1.3 billion tons of food waste, including fruits, vegetables, meat, cereals, and dairy products, are produced in the world each year (Lorenz et al., 2017; Xu et al., 2020b). Meanwhile, many studies have proven that the issue of food waste is becoming increasingly serious due to the explosion of population and the improvement of people's living standards. Such a problem is more prominent in China owing to the huge population base. It is reported that 27% of edible food produced for consumption is wasted or lost annually. Over 69% of food waste occurs in the post-harvest processing and storage stages (Gao et al., 2021; Xue et al., 2021). Fig. 1 illustrates the total output of food waste in various countries. Approximately 60 million tons of food waste was produced in 2010 in China, which took the top spot globally. The United States and the United Kingdom figures were about 35 and 6.7 million tons, respectively. Food waste tends to deteriorate rapidly and can induce many

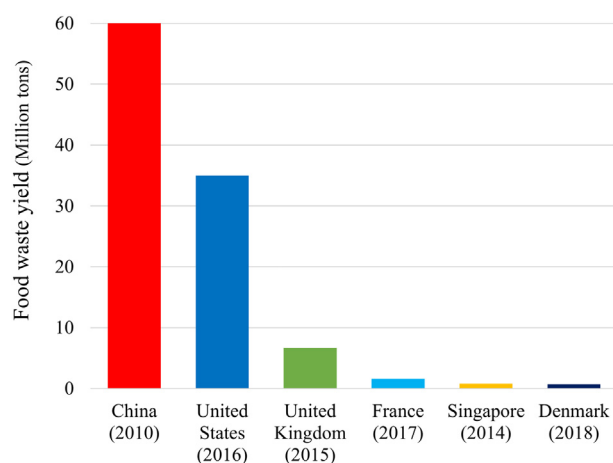


Fig. 1. The output of food waste in different countries (Bellemare et al., 2017; Dou and Toth, 2021; Fei et al., 2021; Jeswani et al., 2021; Magazzino et al., 2021; Mourad, 2016; Tekler et al., 2019).

grave environmental and health problems like bad odor release, harmful gas emission, bacteria leaching, and virus transmission without proper treatment (Yin et al., 2016; Yong et al., 2015).

Consequently, several disposal techniques such as composting, landfilling, animal feeds, and anaerobic digestion, have been developed to deal with food waste, but these methods are difficult to obtain satisfactory outcomes. Composting may turn food waste into fertilizer but needs plenty of land and long reaction time. It will produce malodor, leachate, and toxic gases and easily cause secondary contamination owing to heavy metals and organic matter (Jouhara et al., 2017). Landfilling is the most common disposal method for municipal solid wastes including food wastes. However, this technology is unsustainable and will release the toxic gases, cause water and soil contamination, and occupy massive land (Istrate et al., 2020; Zamri et al., 2021a). The utilization of food waste as animal feed is an old disposal method, which may spread viruses like foot-and-mouth disease and African swine fever (Saleemdeen et al., 2017). Anaerobic digestion is widely used to convert food waste into combustible gas; however, it has drawbacks like poor efficiency, high cost, residual slurry production, and greenhouse gas emission (Lv et al., 2021; Zamri et al., 2021b). Furthermore, other recycling techniques and applications of food waste are also being extensively studied. Xiong et al. (2019) converted food waste into value-added compounds such as acids, sugars, and their derivative products. O'Connor et al. (2021) applied food waste as a biofertilizer and soil-amendment product and emphasized that food waste fertilizer exhibited great potential in improving the properties of soil and increasing agricultural productivity. In addition, researches on food waste-derived renewable solid catalysts and CO₂ adsorbents are also being carried out (Igalavithana et al., 2020; Xiong et al., 2021).

The unceasing deterioration of the environment and the gradual depletion of fossil fuels have stimulated the research on the conversion of food waste into renewable resources (Zhang et al., 2020). Four primary types of thermochemical processes, including incineration, gasification, hydrothermal liquefaction, and pyrolysis, have been utilized in the treatment of food waste to produce heat, electricity, biogas, biochar, bio-oil, and other valuable products. Nevertheless, incineration produces noxious gas and ash; gasification needs high energy input and operational cost, while hydrothermal liquefaction has the drawback of huge capital cost (expensive autoclaves), low security, and unobservable reaction process (Asim et al., 2014; Fonts et al., 2012; Makarichi et al., 2018). As a result, the advantages of pyrolysis, such as the wide range of products, high efficiency, and environmental friendliness, are more apparent. Pyrolysis is the thermal degradation of organic molecules in anaerobic environment by breaking the chemical bonds at a

medium temperature (300–800 °C). The operation conditions, including the temperature, residence time, heating rate, inert gas type and flow rate, have a considerable influence on the pyrolysis process and product distribution and properties (Lee et al., 2020c; Motasemi and Afzal, 2013). Pyrolysis is divided into slow pyrolysis, fast pyrolysis, and flash pyrolysis. In particular, valuable products, biochar and bio-oil, have a wide application in scientific research and industrial production. The adjustment of pyrolysis parameters can optimize the yield of the target product (Wang et al., 2021c). Indisputably, transforming food waste into these valuable products via pyrolysis is a highly viable choice, coinciding with the principle of circular economy (Mak et al., 2020; Mofijur et al., 2021).

Several detailed reviews on the pretreatment methods, main products, relevant laws and regulations, and development prospects of food waste pyrolysis have been published. Elkhaliifa et al. (2019) have summarized recent development work and potential opportunities for biochar production from the pyrolysis of food waste. Karmee (2016) has evaluated the technical feasibility, prospects, and policies for the production of biofuel from food waste through pyrolysis. Yukesh Kannah et al. (2020) have introduced pretreatment methods and products of food waste pyrolysis. Kim et al. (2020b) have presented a review on the pyrolysis of food waste for the generation of biofuel. However, limited work has concentrated on the advanced pyrolysis of food waste, especially co-pyrolysis.

As previously stated, the causes of food waste are varied and complicated, happening at every stage of the supply and consumption chain, with many stakeholders engaged from farm to fork. Knowledge of the nature of food waste, as well as traditional and advanced disposal methods, will provide the groundwork for scientific treatment and sustainable management of food waste. The critical knowledge on the pyrolysis mechanism of food waste, the impact of adding co-feedstocks on product characteristics, and the catalytic effect on the pyrolysis process and product properties is still lacking. Consequently, this article provides a comprehensive review of food waste pyrolysis, with a special emphasis on the advanced pyrolysis of food waste. It introduces the pyrolysis reactor types, provides the characteristics of food waste, summarizes the traditional and advanced pyrolysis of food waste, and analyses the prospects and future challenges of food waste pyrolysis.

2. Characteristics of food waste

As a heterogeneous compound, food waste composition is complex and related to multiple factors, like dietary habits, living styles, climate, region, culture, environment, customs, and history. Table 1 lists the range of the content of each component in food waste according to many studies. Vegetables and fruits, meats, and cereals are the three main components of food waste, containing many organic substances. Meanwhile, carbohydrates, lipids, and proteins are the primary biopolymers in food waste, accounting for 35.1–67.0%, 10.0–35.0%, and 7.0–27.8%, respectively (Chang and Chen, 2010; Chen et al., 2019a). Moreover, the density of food waste is in the interval of 120–480 kg/m³ (Chandrappa and Das, 2012).

Furthermore, the physicochemical characteristics of materials are important for bioenergy recovery, which plays an important part in the pyrolysis process and final products. Those vital properties can be determined with the assistance of elemental and proximate analyses. Table 2 summarizes the elemental and proximate analyses of food

waste, the primary components of food waste, and other materials. The physicochemical characteristics of food waste vary based on the location due to the above reasons. For example, food waste in Annapolis possesses the maximum carbon content and higher heating value (HHV). In contrast, food waste in Auckland has the minimum carbon content and low HHV, but the difference is not significant as a whole. The moisture content of some species is as high as 85.71 wt% due to the high water content of food waste, but after the drying process, the moisture content has dropped to 2.47 wt% or less. Moreover, the carbon content is relatively low, while the oxygen content is high, resulting in a low HHV. The primary components of food waste possess distinct characteristics depending on the type. Meat and peanuts crisps show a favorable pyrolysis potential. Co-feedstocks also display different physicochemical properties. For instance, tire waste has the highest HHV and carbon content, whereas polyvinyl chloride (PVC) possesses the maximum chloride and volatile matter content. In contrast, *Chlorella vulgaris* has the highest nitrogen content owing to the existence of proteins (Al-Lal et al., 2015). Furthermore, the carbon, hydrogen, moisture, and volatile matter content of lignocellulosic biomass (such as wood bark, corn straw, and herbal medicine byproduct) are similar to those of food waste, while the nitrogen content is lower than that of food waste.

3. Conventional pyrolysis of food waste

Pyrolysis is the thermal degradation of organic components into smaller molecules at a moderate temperature in the absence of oxygen. Inert gases, such as nitrogen, argon, helium, and carbon dioxide or vacuum pump, are employed to achieve anaerobic conditions. Table 3 summarizes the result and observation of conventional pyrolysis of food waste.

3.1. Type of reactors

Reactor type significantly influences the pyrolysis process and final products due to the different heat transfer methods, residence time, mixing mode, and pyrolysis efficiency. Fig. 2 depicts the characteristics of various reactors utilized in the pyrolysis of food waste. Due to its convenience and versatility, over half of food waste pyrolysis is carried out through the thermogravimetric analyzer (TGA). Usually, TGA tests are performed along with gas chromatography and mass spectrometer to determine pyrolysis products, which is useful for selecting the optimal raw materials and conditions from various possible substrates and experimental conditions. For example, Giwa et al. (2018) and Opatokun et al. (2015) employed a typical TGA system named the Mettler Toledo Stare System to investigate the pyrolysis characteristics of food waste. It included baffles, reactive gas capillary, gas outlet, temperature sensors, furnace heater, furnace temperature sensor, adjustment ring weights, protective and gas purge protector, and thermostatic balance chamber. Liu et al. (2020a) utilized a simultaneous thermal analyzer to conduct the thermogravimetric analysis, and 5 mg of food waste sample was heated under various conditions. Moreover, Ming et al. (2020) applied a TGA coupled with a Fourier Transform Infrared Spectrometer to explore the released volatile characteristics and thermal decomposition behaviors during the whole pyrolysis process of food waste. However, due to the insufficient sample sizes, the potential sampling bias and poor scalability are the major bottlenecks (Anuar Sharuddin et al., 2016).

About 30% of food waste pyrolysis research is conducted via the fixed-bed reactor because of its simple structure, convenient operation, and technology maturity. Moreover, the fixed-bed reactor can accommodate larger samples than TGA, which possesses more practical significance and reference value. For instance, Chen et al. (2019b) carried out the pyrolysis test in a fixed-bed reactor, including a tube furnace, gas cylinder, thermocouple, condenser, and 5 g sample was injected into a quartz boat. Wang et al. (2020b) also employed a stainless steel fixed-

Table 1

The main components of food waste (Caton et al., 2010; Giwa et al., 2018; Jo et al., 2017; Lee et al., 2017; Liu et al., 2014b; Nagy and Dobó, 2020).

Components	Samples	Content (%)
Vegetables and fruits	Potato, onion, tea, apple, orange, spent coffee	23.08–65.00
Meats	Meat, fish, bones, egg, dairy, poultry, oils	19.00–44.23
Cereals	Rice, wheat, sorghum, corn, millet	16.00–32.69

Table 2
Elemental and proximate analyses of food waste and co-feedstocks.

Biomass	Elemental analysis (dry basis, wt%)					Proximate analysis (wt%)				HHV (MJ/kg)	Reference
	C	H	O	N	S	VM	MO	FC	Ash		
Food waste (location)											
Food waste (Guangzhou, China)	42.34	7.54	33.50	3.68	0.31	81.63	DB	5.74	12.63	18.92	(Liu et al., 2014b)
Food waste (Guangzhou, China)	45.15	8.57	36.31	1.53	0.00	80.27	2.47	16.07	1.19	18.74	(Chen et al., 2018)
Food waste (Nanjing, China)	43.26	8.59	30.25	5.12	–	79.68	DB	10.30	10.02	–	(Zhang et al., 2015)
Food waste (Qingdao, China)	43.38	6.26	37.50	4.49	0.96	66.47	5.10	21.02	7.41	18.91	(Xu et al., 2020a)
Food waste (Yangjiang, China)	40.24	6.83	37.20	5.33	–	74.30	DB	15.30	10.40	–	(Peng et al., 2021)
Food waste digestate (Wuhan, China)	43.52	4.84	39.40	1.92	0.11	66.10	DB	23.69	10.21	–	(Liu et al., 2020a)
Food waste digestate (Sydney, Australia)	42.10	5.20	20.38	5.81	0.91	61.80	DB	12.60	25.60	–	(Opatokun et al., 2015)
Raw food waste (Sydney, Australia)	46.10	5.70	40.79	1.74	0.17	73.40	DB	21.10	5.50	–	(Opatokun et al., 2015)
Food waste (Annapolis, United States)	52.07	7.09	33.01	3.33	0.24	84.90	71.40	12.90	3.34	21.90	(Caton et al., 2010)
Food waste (Auckland, New Zealand)	39.00	7.32	47.70	5.70	0.30	79.10	76.50	14.50	6.40	15.00	(Saqib et al., 2018)
Food waste (Seoul, South Korea)	47.50	6.60	41.60	3.90	0.40	71.30	3.30	15.10	10.30	–	(Park et al., 2020)
Food waste (Kangwon, South Korea)	47.50	12.20	29.70	2.90	–	79.00	85.71	17.25	3.75	19.50	(Jo et al., 2017)
Food waste (Mangalore, India)	46.70	7.28	35.64	2.98	–	70.12	7.60	14.88	7.40	19.87	(Kadlimatti et al., 2019)
Components (location)											
Bone (Beijing, China)	29.17	4.09	20.17	3.29	0.32	52.89	4.82	3.35	38.95	11.90	(Giwa et al., 2018)
Pork (Qingdao, China)	43.56	8.12	33.85	9.75	0.96	83.59	DB	12.56	3.76	19.57	(Ming et al., 2020)
Soybean protein (Linyi, China)	43.90	5.60	30.30	6.10	0.50	71.40	7.90	15.00	5.80	18.23	(Tang et al., 2018)
Meat and bone (Tianjin, China)	37.08	5.45	32.92	5.73	–	70.47	DB	10.74	18.80	–	(Xue et al., 2019)
Fruit pericarp (Tianjin, China)	45.26	6.05	45.65	0.31	–	73.86	DB	23.34	2.81	–	(Xue et al., 2019)
Stems leafy vegetables (Tianjin, China)	42.07	5.60	37.28	3.51	–	68.33	DB	20.22	11.45	–	(Xue et al., 2019)
Bean dreg (Tianjin, China)	49.34	7.90	34.04	4.54	–	80.03	DB	15.81	4.17	–	(Xue et al., 2019)
Meat (Kangwon, South Korea)	54.50	11.30	22.40	10.50	–	86.67	71.27	8.84	4.49	25.20	(Jo et al., 2017)
Vegetables (Kangwon, South Korea)	38.40	6.80	36.30	4.90	–	72.84	95.28	9.40	17.76	14.93	(Jo et al., 2017)
Rice (Qingdao, China)	44.28	7.85	42.99	1.24	0.78	80.42	DB	16.72	2.86	18.04	(Ming et al., 2020)
Cereals (Kangwon, South Korea)	41.40	8.30	47.60	1.10	–	88.95	65.74	10.82	0.23	16.46	(Jo et al., 2017)
Cereals (Ostrava, Czech Republic)	43.00	5.89	41.40	2.16	0.12	73.50	DB	5.50	1.90	19.86	(Grycová et al., 2016)
Peanuts crisps (Ostrava, Czech Republic)	49.80	6.66	34.70	1.93	0.10	79.10	DB	4.60	2.20	22.13	(Grycová et al., 2016)
Co-feedstocks											
Wood bark	51.10	5.80	42.90	0.20	–	67.30	6.60	24.80	1.30	20.68	(Park et al., 2020)
Corn stalk	41.77	5.46	42.34	0.45	–	82.37	7.27	6.30	4.06	–	(Zhang et al., 2015)
Herbal medicine byproduct	46.60	6.10	43.20	1.50	–	80.40	7.70	9.30	2.60	–	(Lee et al., 2021b)
PVC	39.50	4.90	53.20 (Cl)	0.50	1.80	94.70	0.20	5.10	0.04	20.67	(Tang et al., 2018)
Tire waste	82.41	6.48	2.63	0.55	1.67	60.58	–	33.16	6.26	34.60	(Chen et al., 2019b)
<i>Chlorella vulgaris</i>	46.82	8.63	32.03	11.05	0.29	77.23	1.98	12.35	8.44	22.82	(Chen et al., 2018)

Table 3
Summary and observation of conventional pyrolysis of food waste.

Biomass	Conditions	Reactor	Relevant result and observation	Reference
Food waste	700–900 °C, 100–250 mL/min, N ₂ , H ₂	Fixed-bed reactor	- The rise of temperature reduces the bio-oil and biochar yield while increases the biogas yield and LHV. - The rise of carrier gas flow rates increases the biogas yield but decreases the biogas LHV and bio-oil yield. - The rise of H₂ content firstly increases the biogas yield and then reduces it, while the bio-oil yield presents an opposite trend.	(Xu et al., 2020a)
Food waste	800–1200 W, 2 h, CO ₂ /N ₂	Microwave reactor	- The rise of microwave power promotes the formation of biogas. - Pyrolysis under CO₂ atmosphere generates 44.13 wt% more biogas than that under N₂. - Pyrolysis under CO₂ atmosphere improves the specific surface area of biochar.	(Lin et al., 2020)
Food waste	320–480 °C, 30 min, N ₂	Microwave reactor	30.24 wt% bio-oil has been obtained at the conditions of 400 °C, 30 min, and 50 mL/min N₂. - Bio-oil contains alcohols, phenols, carboxylic acid, alkenes, aromatic compounds, aliphatic amines, and nitroalkanes.	(Kadlimatti et al., 2019)
Food waste	300–600 W, N ₂	Microwave reactor	- The rise of microwave power reduces the biochar yield, increases the bio-oil and biogas yield. - The rise of microwave power increases bio-oil production energy but reduces biochar production energy.	(Liu et al., 2014b)
Food waste feedstuff/Food waste compost	300–500 °C, 15–60 °C/min, N ₂	Fixed-bed reactor	- Long retention time promotes the biochar quality. - The rise of temperature lessens the biochar yield. 400 °C is optimal to gain the best quality of biochar.	(Lee et al., 2019)
Raw food waste/Food waste digestate	300–700 °C, 10 or 60 °C/min, N ₂	Fixed-bed reactor	- The rise of temperature decreases the biochar yield. - The rise of temperature significantly increases the large open pores of biochar in raw food waste pyrolysis. - More bio-oil is generated during the pyrolysis of raw food waste. - Bio-oil consists of phenols, esters, hydrocarbons and their derivatives, and the heating rate exerts a huge impact on the bio-oil composition.	(Opatokun et al., 2016)

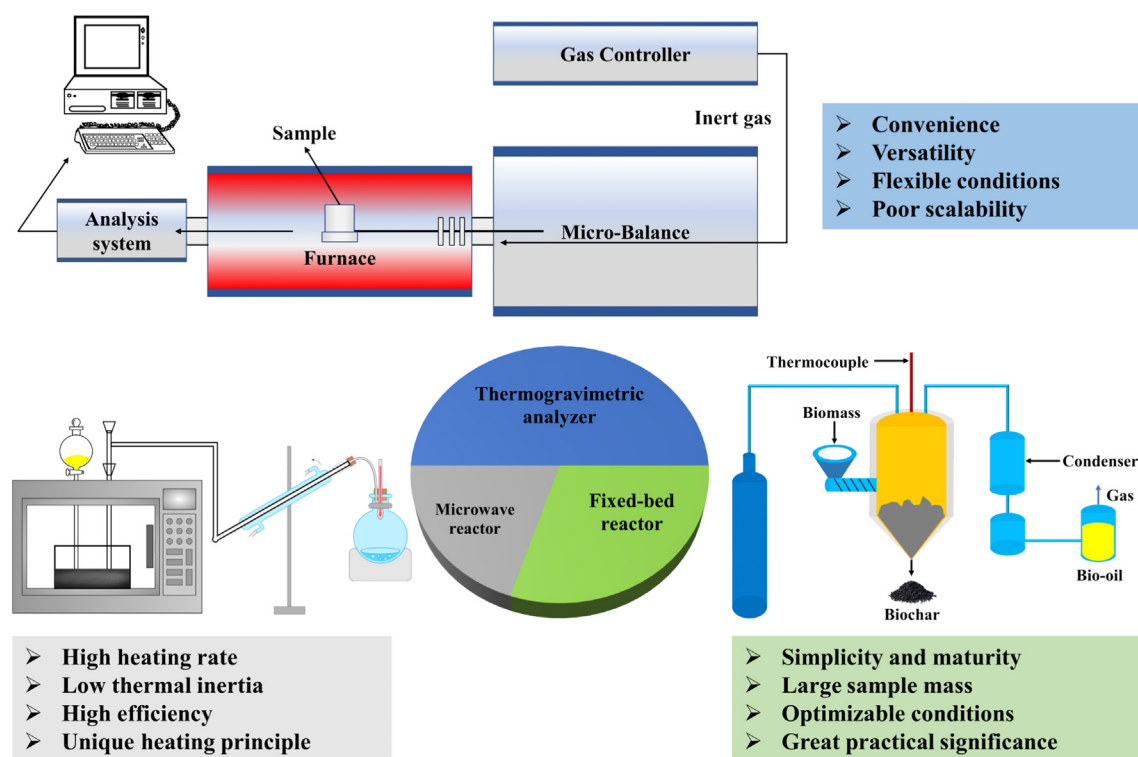


Fig. 2. Characteristics of various reactors (Czajczyńska et al., 2017; Đurić et al., 2014; Mohamed et al., 2016).

bed reactor with 70 mm inner diameter and 140 mm height. It consisted of a gas preheating chamber, reactor feeder, tube furnace, ice-cooled liquid collectors, and gas collection airbags.

Microwave-assisted pyrolysis exhibits a unique heating mechanism in comparison with conventional pyrolysis. Generally, conduction, convection, and radiation are the principal energy transfer methods in conventional heating mode. The sample needs to be in contact with the hot surface to establish a temperature gradient from the outside to the inside of the particle (Wang et al., 2021b). The non-contact volume heating of the microwave employs the polar molecules of the sample to produce heat. When polar molecules try to rearrange themselves under an alternating electric field, energy is lost in the form of heat due to molecular friction and dielectric loss. Microwave penetrates the sample, and microwave energy is directly transmitted to the core of the particle, inducing a temperature gradient from the inside to the outside of the particle, and it possesses the advantages of high heating rate, low thermal inertia, and high efficiency (Fattah et al., 2020; Lee et al., 2020c). The unique heating principle brings a huge difference in the pyrolysis process and final products between both heating modes. Lin et al. (2020) carried out the pyrolysis test via a microwave-assisted pyrolysis system, which contained a temperature control system, multi-mode and linear adjustable microwave oven, and product-collection device. Kadlimatti et al. (2019) chose a microwave-assisted pyrolysis system to pyrolyze food waste, and it consisted of nitrogen supply, microwave cavity, thermocouple, and condenser.

3.2. Product of food waste pyrolysis

Table 4 presents the product distribution of pyrolysis or co-pyrolysis of food waste. The yield of bio-oil from the non-catalytic pyrolysis of food waste is around 60 wt% under the optimal conditions, but the bio-oil quality is not high. Meanwhile, biochar yield is in the range of 24 to 36 wt%, and its HHV is relatively higher than that of bio-oil. Biogas yield is in a wide interval due to the various conditions, while its lower heating value (LHV) is at a low level.

3.2.1. Mechanism of food waste pyrolysis

Carbohydrates, lipids, and proteins are the main organic constituents of food waste. Each one possesses different pyrolysis pathways and products (Liu et al., 2019). As a subclass of carbohydrates, glucose is the most abundant monosaccharide with the molecular formula $C_6H_{12}O_6$. Dehydration and reduction are the primary chemical reactions during the pyrolysis of glucose. Dehydration reaction converts glucose into anhydro-sugars, the reduction of glucose generates sorbitol, and the dehydration reaction transforms sorbitol into 1,4-sorbitan, then further converts 1,4-sorbitan into isosorbide (Carlson et al., 2010). Triglycerides are the major component of lipids. A series of reactions, including decarbonylation, decarboxylation, and fragmentation, take place and transform glycerol into hydrocarbons, esters, acids, alcohols, and alicyclic compounds. Moreover, the degradation of steroids also generates steroid-derived hydrocarbons (Wang et al., 2017b). Proteins are composed of amino acids, connected by peptide bonds. Amine and diketopiperazine species are the main products from the pyrolysis of proteins. Amine species are formed by breaking the bonds of the protein structure. While diketopiperazine species are produced by the cyclization of adjacent amino acids contained in polypeptides, those amino acids are subsequently cleaved. Meanwhile, many types of pyrrolidines are generated during the production of diketopiperazine species (Orsini et al., 2017).

In addition, other regular materials present distinct pyrolysis pathways. For plastics, the dissociation of the C—C bond is the primary reaction of polymer thermal decomposition, along with some secondary reactions, such as unsaturation, isomerization, cyclization, recombination, aromatization, diene synthesis, and further chain scission to form cycloalkanes and aromatic hydrocarbons (Cheng et al., 2020; Sadrameli, 2015). Tire contains natural rubber, butadiene rubber, isoprene rubber, and styrene-butadiene rubber. Each material possesses different pyrolysis pathways and products. Free radicals in natural rubber experience rearrangement and cyclization to produce α -limonene or free radicals from the conjugate system to generate isoprene. Diels-Alder reaction converts isoprene into α -limonene. For butadiene

Table 4
Product distribution of food waste pyrolysis or co-pyrolysis.

Biomass	Catalysts	Conditions	Reactor	Liquid products		Solid products		Gas products		Reference
				Yields (wt%)	HHV (MJ/kg)	Yields (wt%)	HHV (MJ/kg)	Yields (wt%)	LHV (MJ/Nm ³)	
Pyrolysis										
Food waste	–	700 °C, N ₂	Fixed-bed reactor	29.86	–	32.99	–	37.15	10.91	(Xu et al., 2020a)
Raw food waste	–	500 °C, 10 °C/min, N ₂	Fixed-bed reactor	60.30	11.20	32.30	24.80	7.40	15.70	(Opatokun et al., 2016)
Food waste digestate				52.20	13.50	42.50	13.00	5.30	17.20	
Cereals	–	800 °C, 10 s, N ₂	Fixed-bed reactor	62.00	27.14	24.00	31.44	14.00	–	(Grycová et al., 2016)
Peanuts crisps				46.00	14.10	26.00	29.60	28.00	–	
Spent coffee waste	–	400 °C, 100 kg/h, N ₂	Fluidized-bed reactor	55.50	27.91	28.12	–	16.38	–	(Vu Ly et al., 2022)
	–	460 °C, 100 kg/h, N ₂		59.43	35.42	22.32	–	18.25	–	
	–	500 °C, 100 kg/h, N ₂		52.52	37.80	21.85	–	25.63	–	
	Dolomite			54.77	37.54	23.62	–	21.62	–	
	HZSM-5			50.10	40.46	25.65	–	24.25	–	
	Hematite			54.04	38.99	17.44	–	28.52	–	
	Magnetite			51.64	36.96	23.41	–	24.95	–	
Food waste	–	300 W, N ₂	Microwave reactor	36.00	32.91	36.00	18.23	28.00	–	(Liu et al., 2014b)
	CaO			13.00	32.16	79.00	20.58	8.00	–	
	MgO			30.00	32.83	35.00	15.45	35.00	–	
	CuO			13.50	31.96	80.00	21.40	6.50	–	
	Fe ₂ O ₃			29.50	33.25	38.00	17.53	32.50	–	
	MnO ₂			26.00	32.46	35.00	13.90	39.00	–	
	CuCl ₂			20.00	32.84	39.50	20.38	40.50	–	
	NaCl			27.00	33.05	41.00	18.48	32.00	–	
	MgCl ₂			9.50	32.48	78.50	22.43	12.00	–	
Co-pyrolysis										
Food waste/ <i>Chlorella vulgaris</i>	–	500 °C, 25 min, CO ₂	Fixed-bed reactor	35.00	30.00	39.00	–	26.00	–	(Sajjadi et al., 2018)
		600 °C, 25 min, CO ₂		30.00	33.00	46.00	–	24.00	–	
		700 °C, 25 min, CO ₂		25.50	27.00	52.00	–	22.50	–	
Food waste/PVC	–	300 °C, N ₂	Fixed-bed reactor	32.00	–	41.00	–	27.00	–	(Peng et al., 2021)

rubber, free radicals are transformed into 1,3-butadienes by scission and dehydrogenation reactions; 4-vinyl-1-cyclohexenes are produced with the rearrangement and cyclization of free radicals; 1,3-cyclopentadienes are formed by dehydrogenation and cyclization reactions; the other product 1,4-cycloheptadienes are generated by the cyclization. Free radicals derived from styrene-butadiene rubber are cracked into benzene derivatives through scission and dehydrogenation reactions (Li et al., 2020).

Lignocellulosic biomass consists of hemicellulose, cellulose, and lignin. Levoglucosan originates from the decomposition of cellulose generates furan via decarboxylation, dehydration, or decarbonylation reactions. Afterwards, olefins and aromatic compounds are formed through the decarbonylation and oligomerization of furan (Wang et al., 2014). The pyrolysis of hemicellulose generates double-hydrated xylose, and some products are similar to those from the pyrolysis of cellulose. Lignin exhibits unique pyrolysis pathways and products. Monomeric phenolic compounds are formed from the pyrolysis of lignin. The resulting product generates phenolic compounds and polycyclic aromatic hydrocarbons through oligomerization and aromatization (Wang et al., 2017a).

Table 5 summarized the properties of products that originated from the pyrolysis of various materials, such as food waste, lignocellulosic

biomass, microalgae, plastics, tire, sewage sludge, animal manure, and medical waste. Different components and physicochemical properties of those materials exert a distinct effect on the product properties. Generally, materials with high ash content, including animal manure and sewage sludge, are detrimental to the formation of bio-oil but promote the generation of biochar. While materials with high volatile matter content, like plastics and medical waste, are conducive to the production of bio-oil. Moreover, feedstocks with high effective hydrogen index (EHI), such as plastics, tires, and medical waste, can significantly improve the quality of pyrolysis products. As for food waste, sole pyrolysis of food waste displays an ordinary performance due to the moderate levels of volatile matter and EHI.

3.2.2. Effect of operating parameters

During the pyrolysis process, food waste is heated to a preset temperature exceeding the thermal stability threshold in the absence of oxygen and undergoes various chemical reactions like decarboxylation, dehydration, fragmentation, and dehydrogenation to produce a series of functional groups. Heavy components are converted into biochar, some volatile matters are recondensed into bio-oil, and non-condensable gases are collected or directly released into the air (Sharifzadeh et al., 2019). Therefore, pyrolysis parameters play a

Table 5
The properties of products from the pyrolysis of food waste and other materials.

Biomass	Liquid products		Solid products		Gas products		Reference
	Yields (wt%)	HHV (MJ/kg)	Yields (wt%)	HHV (MJ/kg)	Yields (wt%)	LHV (MJ/Nm ³)	
Food waste	29.86–62.00	11.20–32.91	24.00–42.50	13.00–31.44	5.30–37.15	10.91–17.20	This study
Lignocellulosic biomass	8.00–69.90	5.01–34.40	15.20–80.00	18.20–33.94	12.00–60.80	13.40–17.30	(Dhyani and Bhaskar, 2018; Huang et al., 2016; Santos et al., 2020)
Microalgae	20.00–55.00	26.80–45.90	15.00–39.70	14.50–36.67	3.90–57.30	4.60–12.90	(Lee et al., 2020a; Yang et al., 2019)
Plastics	12.30–96.73	21.10–43.00	0.00–28.13	18.84–36.29	2.50–76.90	42.00–50.00	(Anuar Sharuddin et al., 2016; Maqsood et al., 2021)
Tire	8.70–65.40	38.00–42.10	32.90–52.30	29.30–31.50	2.00–35.20	29.90–42.10	(Arabourrutia et al., 2020; Williams, 2013; Zhang et al., 2021)
Sewage sludge	20.10–39.90	18.80–36.40	19.90–56.00	11.00–15.00	13.60–18.60	9.94–24.70	(Gao et al., 2020; Liu et al., 2021)
Animal manure	17.00–41.50	6.50–29.00	8.50–62.00	14.00–35.50	18.00–49.19	6.00–17.00	(Erdogdu et al., 2019; Jeong et al., 2015; Mong et al., 2020a)
Medical waste	42.00–83.30	37.56–42.54	0.00–50.69	22.80–33.75	2.50–43.00	39.16–46.18	(Luo et al., 2021; Su et al., 2021)

significant role in the optimization of the target product yield and quality. Tailoring the pyrolysis parameters influences the properties of bio-oil, biochar, and biogas greatly during the pyrolysis of food waste.

Firstly, pyrolysis conditions, especially temperature and microwave power, have a tremendous effect on the product distribution since they directly influence the breaking of chemical bonds in biomass. Xu et al. (2020a) pyrolyzed food waste to realize the target of the Reduce-Recovery-Recycle strategy and found that the pyrolysis temperature largely affected the product distribution. The bio-oil yield experienced a steady decline, while the biogas yield continually increased with the rise of temperature, but there was no significant change in the biochar yield. This was because that high temperature promoted the cracking of condensable gas into non-condensable gas.

Moreover, the high microwave power can reach the desired temperature within a short time, which indicates that high microwave power is equivalent to a high heating rate. Fig. 3 presents the relationship between microwave power and product distribution during the pyrolysis of food waste. The bio-oil yield increased firstly and then descended. The biogas yield rose gradually, while the biochar yield showed a steady downward trend when the microwave power rose (Lin et al., 2020). Liu et al. (2014b) also detected that the rise of microwave power greatly influenced the product distribution during the pyrolysis of food waste, which reduced the biochar yield and raised the biogas yield continuously. The result was also supported by Hossain et al. (2016) and Fan et al. (2019). In addition, Liu et al. (2020a) observed that the growth of heating rate led to the shifts in TG and DTG curves. Ming et al. (2020) also detected the same phenomenon and discovered that the increment of heating rate reduced the mass of residues because food waste decomposed quickly and produced higher volatiles under high heating rates.

Furthermore, the carrier gas is used to transport vaporized products. A variety of inert gases, including nitrogen, argon, helium, and carbon dioxide, have been applied in pyrolysis, each carrier gas possesses distinct reactivity due to its unique molecular weight. Kim and Lee (2020) employed N₂ and CO₂ as fluidizing gas during the pyrolysis of food waste. They observed that the utilization of CO₂ impeded the formation of condensable gas and promoted the generation of non-condensable gas, but it had a limited effect on the biochar yield. Moreover, the cyclic compounds in bio-oil experienced a decline with the usage of CO₂. The presence of CO₂ induced the Boudouard reaction, which enhanced the production of CO at a high temperature (720 °C); CO₂ shifted the carbon distribution from condensable to non-condensable gas. Similarly, as depicted in Fig. 3, Lin et al. (2020)

conducted the microwave pyrolysis of food waste under N₂ and CO₂ and also found that more biogas was obtained under the CO₂ atmosphere because more microwave energy was absorbed in the CO₂ environment, which enhanced the decomposition of bio-oil. In addition, the presence of CO₂ improved the biogas quality by increasing the content of CO and raised the specific surface area of biochar to 205.71 m²/g.

3.2.3. Bio-oil

Bio-oil originated from conventional pyrolysis of food waste comprises aliphatic and aromatic hydrocarbons, oxygenated compounds including phenols, aldehydes, ketones, esters, acids, and alcohols, and nitrogenous compounds, such as nitriles and amides. Meanwhile, pyrolysis conditions like temperature and atmosphere significantly impact the composition and quality of bio-oil. It is reported that aliphatic, phenol, and carbonyl show a remarkable decline, but aromatics and nitrogenous compounds present an upward trend as the temperature increases. The underlying cause is that an elevated temperature is conducive to the chain-scission, cyclization, aromatization, and isomerization reactions, which converts aliphatic into aromatics. Moreover, the deoxygenation, dehydroxylation, aromatization, and polymerization of phenols is prone to occur, and functional groups like -C-O-, -OH-, -C=O-, and -COOH from alcohols and carbonyls are very unstable under a high temperature (800 °C) (Xu et al., 2020a). Lin et al. (2020) pyrolyzed the food waste under the CO₂ atmosphere and explored the positive effects of CO₂ on the pyrolysis process. Polycyclic aromatic hydrocarbons (PAHs), aliphatic hydrocarbons, oxygenated compounds, and mononuclear aromatics decreased steadily, while nitrogenous compounds continuously rose from 9.14 to 22.34 wt% with the increase of CO₂ content. This was ascribed to CO₂ enhanced the decomposition of oxygenated compounds and hydrocarbons to generate small molecule fragments and gas. Those molecule fragments reacted with straight-chain nitrogenous compounds to produce nitrobenzene derivatives via the cyclization and aromatization reactions. Kadlimatti et al. (2019) carried out the experiment of food waste pyrolysis and detected the physical and chemical properties of bio-oil. The pH, density, and viscosity were 4.5, 1046.36 kg/m³, and 1.16 cSt, respectively, while the bio-oil HHV was as low as 19.95 MJ/kg, which was difficult to replace gasoline for commercial use.

The low HHV and yield, high acidity and viscosity hamper the application of food waste-based bio-oil as a commercial fuel. Based on the aforementioned issues, pretreatment techniques including separation, drying, torrefaction, and milling are supposed to be performed to decrease the moisture content and enhance the grinding and hydrophobicity of food waste, thus improving its pyrolysis performance. While the configuration of pyrolysis conditions, such as temperature, residence time, heating rate, inert gas flow rate, and pressure, requires further study to optimize the production of bio-oil. Meanwhile, the research on the large-scale production of food waste-based bio-oil needs more efforts to lay the foundation for the commercialization and industrialization of food waste pyrolysis.

3.2.4. Biochar

Biochar is the solid residue in a char form, which contains carbon, inorganic materials, metals, and transition elements. In general, a high temperature (>700 °C) is not favorable to the formation of biochar. Mild conditions of low temperature, slow heating rate, and long residence time are instrumental in the production of biochar (Xiong et al., 2020).

Table 6 illustrates the properties of biochar from food waste pyrolysis or co-pyrolysis. The quality of biochar from sole food waste pyrolysis is not at a high level, as the carbon content is in a wide interval of 37.21 to 90.40 wt%, and the biochar HHV is in the range of 13.42 to 31.44 MJ/kg. However, the solid product originated from the pyrolysis of food waste digestate displays a poor quality (13.00 MJ/kg) in consequence of the bioenergy loss during the digestion process (Opatokun et al.,

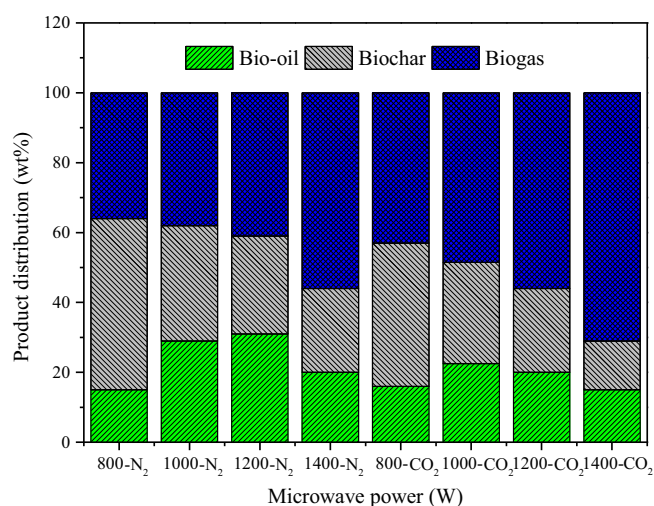


Fig. 3. The effects of microwave power and inert gas types on the product distribution (Lin et al., 2020).

Table 6

Elemental and proximate analyses of biochar from food waste pyrolysis or co-pyrolysis.

Biomass	Conditions	Reactor	Elemental analysis (dry basis, wt%)					Proximate analysis (wt%)				HHV (MJ/kg)	Reference
			C	H	O	N	S	VM	MO	FC	Ash		
Food waste digestate	400 °C, 120 min, N ₂	Fixed-bed reactor	46.89	2.12	14.66	1.90	0.16	36.24	–	29.49	34.27	–	(Liu et al., 2020a)
	600 °C, 120 min, N ₂		50.41	1.26	9.44	1.86	0.09	16.45	–	36.94	46.61	–	
	800 °C, 120 min, N ₂		50.19	1.09	8.92	1.79	0.09	13.83	–	48.25	37.92	–	
Food waste	1000 W, N ₂	Microwave reactor	90.40	1.36	4.20	2.82	1.22	–	–	–	–	–	(Lin et al., 2020)
	1000 W, CO ₂		73.66	1.08	22.00	2.05	1.21	–	–	–	–	–	
	400 °C, 15 min, N ₂		51.42	6.78	25.00	4.20	–	72.70	0.61	12.59	14.10	–	
	400 °C, 30 min, N ₂		54.08	5.75	15.15	4.98	–	62.86	0.37	20.04	16.73	–	
	400 °C, 45 min, N ₂		61.16	5.88	11.14	5.11	–	5.363	0.79	25.40	20.18	–	
Food waste feedstuff	300 °C, 60 min, N ₂	Fixed-bed reactor	57.86	6.35	17.77	5.33	–	64.58	0.60	19.21	15.61	25.84	(Lee et al., 2019)
	400 °C, 60 min, N ₂		61.35	5.87	12.05	5.49	–	50.79	2.76	25.62	20.83	27.09	
	500 °C, 60 min, N ₂		37.21	1.37	11.45	2.89	–	24.60	0.69	41.14	33.57	13.42	
	400 °C, 15 min, N ₂		51.42	6.78	25.00	4.20	–	72.70	0.61	12.59	14.10	23.35	
	400 °C, 30 min, N ₂		54.08	5.75	15.15	4.98	–	62.86	0.37	20.04	16.73	24.09	
	400 °C, 45 min, N ₂		61.16	5.88	11.14	5.11	–	5.363	0.79	25.40	20.18	27.13	
	500 °C, 4 h, Ar		71.30	2.10	6.44	2.64	0.12	13.30	–	69.30	17.40	24.80	
Raw food waste		Fixed-bed reactor	35.30	1.20	3.86	4.01	0.53	12.60	–	32.30	55.10	13.00	(Opatokun et al., 2015)
Food waste digestate			84.30	0.51	2.50	3.07	0.00	–	–	–	–	31.44	
Cereals	800 °C, 10 s, N ₂	Fixed-bed reactor	80.00	0.35	4.20	2.10	0.00	–	–	–	–	29.60	(Grycová et al., 2016)
Peanuts crisps			19.15	0.91	2.03	2.61	2.12	–	–	–	73.18	–	
Food waste/Sewage sludge	550 °C, 1 h, N ₂	Fixed-bed reactor	72.10	1.67	–	0.33	0.01	–	–	–	–	–	(Wang et al., 2021a)
Food waste/oak (2:1)	700 °C, 15 g/h, N ₂	Fixed-bed reactor	61.70	2.09	–	1.01	0.01	–	–	–	–	–	(Nagy and Dobó, 2020)
	900 °C, 15 g/h, N ₂												

2015). Liu et al. (2020a) conducted the pyrolysis of food waste digestate and found that temperature had an inhibitory effect on the biochar yield but had a promotion effect on the biochar quality, which increased the carbon content from 46.89 to 50.19 wt% and reduced the oxygen content from 14.66 to 8.92 wt% when the temperature rose from 400 to 800 °C. In addition, an elevated temperature decreased the volatile matter content and increased the ash and fix carbon content greatly. Nevertheless, Lee et al. (2019) carried out an experiment of the pyrolysis of food waste feedstuff and obtained a dissimilar result. Table 6 shows the influence of temperature on the biochar quality, including element composition and HHV. The carbon, hydrogen content, and biochar HHV presented an upward trend at first and then declined with the increasing temperature. The optimal temperature was found to be 400 °C to gain the biochar with the maximum HHV of 27.09 MJ/kg. The main reason behind those two different results may be the distinct types of reactors and sample dosages. Liu et al. (2020a) employed the TGA with only 5 mg sample to explore the relationship between temperature and biochar properties. Lee et al. (2019) conducted the experiment in a fixed-bed reactor and obtained 0.7–1.0 g biochar. The conclusion based on the TGA and tiny sample mass has no representativeness. Additionally, many studies also demonstrated that the optimal temperature was around 400 °C to obtain the biochar with an admirable quality, low temperature is detrimental to the degradation of food waste into the volatile matter, resulting in the charring process (Atienza-Martínez et al., 2020; Cantrell et al., 2012; Yildiz et al., 2019). Meanwhile, residence time has a certain effect on the biochar properties, the long residence time is conducive to the increase of carbon content and the decrease of oxygen content, leading to a higher HHV, 45 min is the best from the point of biochar quality and energy consumption (Lee et al., 2019).

Moreover, some studies have introduced CO₂ as a reactive medium to replace N₂ to influence biochar quality. Lin et al. (2020) reported the biochar generated from microwave-assisted pyrolysis of food waste at high temperatures (600–900 °C) under CO₂ environment possessed a more favorable specific surface area and mesoporous than that under N₂ atmosphere. The principal cause was that CO₂ reacted with the carbon to produce CO at a high temperature (700 °C), massive biogas was formed inside the biochar, and the biogas diffusion resulted in the improvement of biochar specific surface. However, the quality of biochar experienced a decline since the carbon content decreased from

90.40 to 73.66 wt% because CO₂ induced the further thermal decomposition of biochar.

As another important product of food waste pyrolysis, biochar can be applied as a catalyst, adsorbent, soil conditioner, and supercapacitor. Biochar is also a carbon sequestration agent, which can effectively curb global warming and its consequences (Wang et al., 2020a). The physical and chemical characteristics of biochar, such as surface area, porosity, nutrient content, and metal content, are strongly related to its performance in a variety of fields. Thus, further research into biochar with particular functions is required, which will broaden its application range and enhance its economic worth. Meanwhile, pyrolysis parameters should be tailored and optimized to ameliorate the characteristics of biochar for a particular application during the pyrolysis of food waste.

3.2.5. Biogas

Approximately 20 wt% biogas is produced during the conventional pyrolysis of food waste, but the biogas quality is low due to the low EHI of food waste. Generally, elevated temperatures and long residence times are conducive to the formation of biogas. Xu et al. (2020a) performed the pyrolysis of food waste and determined the influence of temperature, carrier gas flow rate, and hydrogen content on the biogas composition and LHV. The rise of temperature improved the H₂, CH₄, and C₂₊ content but inhibited the production of CO₂. The main reason for the phenomenon was that elevated temperature enhanced the function of free radical reaction, resulted in the promotion of the production of H₂, CH₄, and C₂₊. Moreover, as presented in Eq. (1), the Boudouard reaction was enhanced under a high temperature (700 °C) to generate CO, which improved the biogas LHV from 10.91 to 14.62 MJ/Nm³.



The increment of carrier gas flow rate had an inhibitory effect on the biogas quality, which reduced the H₂ and CH₄ content but increased the CO₂ content, leading to a low LHV. The high flow rate was similar to the short retention time, which hindered the secondary reaction of volatiles and lowered the biogas quality. Moreover, they employed H₂ as a carrier gas with N₂ to improve the quality of final products. H₂, CH₄, and C₂₊ saw a steady rise with the growth of H₂ content in the carrier gas. Systematic or orderly cracking of chemical bonds of the macromolecules

was induced in the H₂ atmosphere during pyrolysis, and random chain-scission of macromolecules was substituted (Singh et al., 2016). Biogas has a low economic and application value, it can be reused as an energy source in the pretreatment and pyrolysis processes, lowering energy input and transportation costs.

4. Advanced pyrolysis of food waste

Co-pyrolysis and catalytic pyrolysis are the most extensively utilized advanced pyrolysis techniques presently. Both of them possess a positive influence on the whole pyrolysis process and final products with little or no modification of existing equipment (Abnisa and Wan Daud, 2014; Ubando et al., 2020). Many researchers have devoted themselves to related studies and made fruitful results.

4.1. Co-pyrolysis of food waste

Co-pyrolysis has two overriding characteristics: simplicity and effectiveness. The interaction between the volatile matter from different materials and the resulting synergistic effects positively impact the pyrolysis process and product properties. As mentioned above, food waste mainly consists of carbohydrates, lipids, and proteins; lignocellulosic biomass contains hemicellulose, cellulose, and lignin; the tire is composed of rubber, carbon black, and fabric, while plastics are made of polymers. Consequently, adding various materials can bring more new changes and provide more possibilities for product optimization. Sajjadi et al. (2018) carried out the co-pyrolysis of *Chlorella vulgaris* with food waste and detected synergistic effects that increased the hydrocarbons content and decreased the content of the carboxylic acid

largely in bio-oil. The interaction between the volatile matter from two materials played the role of activator, effectively breaking the oxygen-containing chemical bonds.

Usually, the bio-oil yield has a positive correlation to the volatile matter content. The addition of materials with high volatile matter content is beneficial to the production of bio-oil, while co-feedstocks with high ash content hinder the process but facilitate the generation of bio-char and biogas (Dhyani and Bhaskar, 2018; Omar et al., 2011).

Table 7 summarizes the relevant result and observations of food waste co-pyrolysis. Firstly, the addition of co-feedstocks and subsequent synergistic effects have a tremendous impact on the properties of bio-oil. Chen et al. (2019b) observed remarkable synergistic effects on the product distribution and bio-oil quality through the co-pyrolysis of food waste and *Chlorella vulgaris*. Fig. 4 presents the variation of product distribution and bio-oil HHV under different microalgae ratios. The bio-oil yield showed a visible decline to 28.4 wt% firstly, then increased. The biogas yield displayed an opposite trend with the rise of the microalgae ratio. This was mainly attributed to the rising hybridization degree that promoted the decomposition of condensable gas into non-condensable gas. Meanwhile, synergistic effects had a promotional effect on the bio-oil quality, which increased the bio-oil HHV to 33.9 MJ/kg. Furthermore, the addition of *Chlorella vulgaris* influenced the bio-oil composition significantly. Massive nitrogenous compounds, including azines, indoles, pyridines, azoles, nitriles, amides, and amines, were detected in the bio-oil generated from the pyrolysis of microalgae in consequence of the existence of the proteins. As for the bio-oil originated from food waste, acids like octanoic acid, hexadecenoic acid, and octadecenoic acid were the major constituents, which came from the pyrolysis of triglyceride and the thermal degradation of hemicellulose and cellulose in

Table 7
Summary and observation of food waste co-pyrolysis.

Biomass	Co-feedstocks	Conditions	Reactor	Relevant result and observation	Reference
Food waste	<i>Chlorella vulgaris</i>	900 °C, 20–50 °C/min, N ₂	TGA	- Co-pyrolysis improves the thermal gas release and decomposition behavior. - Co-pyrolysis of 50% food waste and 50% microalgae decreases the apparent activation energy by 14.9–28.2%. - The addition of 50% microalgae increases the hydrocarbons content and lessens the saccharides content greatly.	(Chen et al., 2018)
Food waste	<i>Chlorella vulgaris</i>	500–700 °C, 25 min, CO ₂	Fixed-bed reactor	- Apparent activation energy presents the highest value during the co-pyrolysis of 50% food waste and 50% microalgae. - Co-pyrolysis of 50% food waste and 50% microalgae obtains the bio-oil with the highest HHV of 33.9 MJ/kg. - The rise of the microalgae ratio increases the bio-oil and biochar yield while reduces the biogas yield. - The rise of the microalgae ratio increases the hydrocarbons content and reduces the acids content greatly.	(Sajjadi et al., 2018)
Food waste	Tire waste	900 °C, N ₂	TGA	- Co-pyrolysis of 50% food waste and 50% tire waste reduces apparent activation energy by 16.78% via the Ozawa-Flynn-Wall model. - Co-pyrolysis enhances the hydrocarbon content and inhibits the non-hydrocarbon content. - The rise of the tire waste ratio increases the olefins content but hinders the production of PAHs and oxygenated compounds.	(Chen et al., 2019b)
Food waste	Wood bark	300–700 °C, N ₂	Continuous-flow reactor	- The rise of the wood bark ratio increases the biochar yield in most cases but lessens the biochar HHV. - The rise of the wood bark ratio facilitates the formation of phenolic compounds and hinders the production of acids. - The rise of temperature reduces the biochar yield continuously. - Co-pyrolysis at proper ratio favors the production of H₂ and hinders the generation of harmful products like phenols and PAHs.	(Park et al., 2020)
Food waste	Herbal medicine byproduct	300–700 °C, N ₂	Continuous-flow reactor	- Co-pyrolysis of 20% food waste and 80% herbal medicine byproduct gains the maximum H₂ yield at 700 °C. - The rise of the herbal medicine byproduct ratio decreases the biochar yield and HHV. - The rise of the herbal medicine byproduct ratio promotes the formation of phenolic compounds and PAHs.	(Lee et al., 2021b)
Food waste	Corn stalk	500–700 °C, 20s, He	CDS Pyroprobe 5200 pyrolyzer	600 °C is the best for the production of aromatics during co-pyrolysis. - Synergistic effects facilitate the formation of aromatics and other petrochemicals greatly.	(Zhang et al., 2015)
Soybean protein	PVC	400–600 °C, N ₂	Fixed-bed reactor	- Temperature possesses a huge effect on the synergistic effects. - Synergistic effects have a negative impact on the formation of bio-oil but positively affect the production of biochar in 500 °C. - Co-pyrolysis inhibits the formation of nitrogenous compounds except for indole-N.	(Tang et al., 2018)

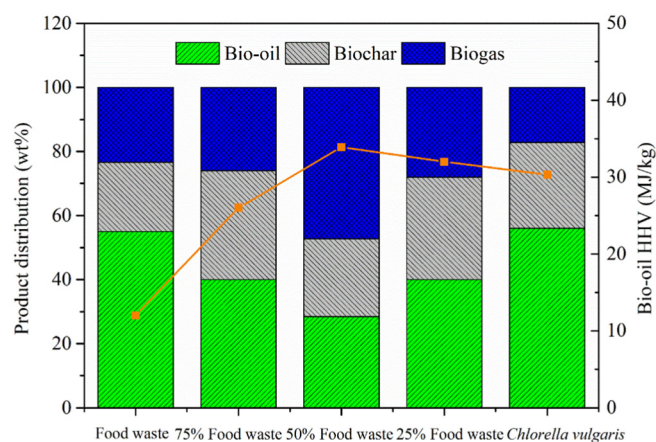


Fig. 4. The effect of microalgae ratio on the product distribution and bio-oil HHV (Sajjadi et al., 2018).

food waste. Moreover, a notable interaction effect was occurred after adding microalgae, which increased the hydrocarbons content and reduced the acids content sharply. This was ascribed to the strong synergistic effects that acted as an activator and broke the oxygenated chemical bonds.

Chen et al. (2019b) chose tire waste as co-feedstock for the food waste pyrolysis and determined the change in the bio-oil composition. Fig. 5 exhibits the variation of the bio-oil composition with the rise of tire waste. Adding 25% tire waste raised the olefins content from 6.80 to 23.30 wt% and reduced the oxygenated compounds and PAHs from 12.52 to 2.70 wt% and 17.00 to 10.80 wt%, respectively. The accession of tire waste with a high H/C ratio provided hydrogen to food waste, enhancing the conversion of non-hydrocarbon to hydrocarbon via hydrogen transfer reactions. Notably, *D*-limonene, a valuable chemical in the olefins, reached the maximum value because synergistic effects accelerated the conversion of isoprene to *D*-limonene through the Diels-Alder reaction. Based on the output of hydrocarbons products, 75% tire waste was optimal for the enhancement of bio-oil quality during co-pyrolysis.

Simultaneously, many reports have demonstrated that synergistic effects possess a huge impact on the properties of biochar. Tang et al. (2018) selected PVC as a co-feedstock for the pyrolysis of food waste and observed that synergistic effects positively influenced the biochar yield and negatively affected the bio-oil yield since the experimental value of biochar was 31% higher than its calculation value, but that of bio-oil was 21% less than the corresponding calculation value. HCl originated from PVC acted as an acid-catalyst to hinder depolymerization

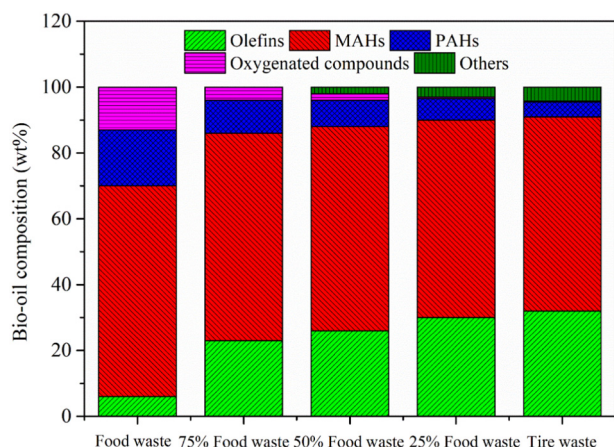


Fig. 5. The effect of tire waste ratio on the bio-oil composition (Chen et al., 2019b).

and promote dehydration and carbonization progress. Moreover, positive synergistic effects were also observed on the reduction of nitrogenous compounds in the bio-oil, as the experimental value of nitrogenous compounds was 77% less than its calculated value due to the disappearance of amine-N, pyridine-N, pyrrole-N, and nitrile-N, only indole-N was detectable in the bio-oil. The primary reason was that the HCl inhibited the production of nitrogenous compounds except for indole-N. Park et al. (2020) pyrolyzed food waste with wood bark and determined that the addition of 50% wood bark increased the biochar yield but decreased the biochar HHV from 23.80 to 22.70 MJ/kg. Moreover, co-pyrolysis was conducive to enhance the specific surface area of biochar, raising the specific surface area and pore volume from 0.278 to 0.44 m²/g and 0.00099 to 0.016 cm³/g after adding 75% wood bark, respectively.

The synergistic effect caused by co-pyrolysis also influences the properties of biogas. Lee et al. (2021b) pyrolyzed food waste with oriental herbal medicine byproducts and found that co-pyrolysis obtained more H₂ than the sole pyrolysis of each raw material. This was because synergistic effects promoted dehydrogenation in the secondary pyrolysis stage (Shen et al., 2017). Park et al. (2020) observed that the application of wood bark in food waste pyrolysis significantly affected the biogas composition and LHV. CO₂ was the principal product in the biogas, alkanes and alkenes showed a downward trend with the increase of wood bark proportion, while H₂ reached the highest yield with 50% wood bark, leading to the maximum LHV of 8.00 MJ/kg.

In addition, co-pyrolysis is instrumental in reducing the apparent activation energy and energy conservation, Chen et al. (2018) observed that the application of microalgae in the pyrolysis of food waste decreased the apparent activation energy by 14.9–28.2%. Chen et al. (2019b) detected that apparent activation energy was reduced by 16.78% after adding 50% tire waste in the pyrolysis of food waste.

However, correlational studies on the co-pyrolysis of food waste with other materials with high EHI such as LDPE, PP, medical waste, and waste oils are relatively deficient and have not been explored as co-feedstocks. The application of those materials as co-feedstocks in the pyrolysis of food waste will further ameliorate the properties of the final product (Zhang et al., 2014).

4.2. Catalytic pyrolysis of food waste

Catalytic pyrolysis is a practical approach to improve the existing pyrolysis process and final product properties, such as enhancing the product quality, shortening the reaction time, optimizing the product distributions, and ameliorating the compatibility of bio-oil with engines (Di Serio et al., 2008). Moreover, according to the stage of adding the catalyst, it can be divided into in-situ (during the pyrolysis) and ex-situ (after pyrolysis and react with volatiles) mode. Table 8 summarizes the catalytic effects on food waste pyrolysis. Several studies have indicated that catalysts possess a promotional effect on the pyrolysis of food waste.

Firstly, the addition of catalysts influences product distribution significantly during the pyrolysis of food waste. Liu et al. (2014b) conducted the pyrolysis of food waste by microwave heating. Fig. 6 depicts the catalytic effect on the product distribution and biochar LHV. The application of CaO, CuO, and MgCl₂ had an inhibitory effect on the formation of bio-oil but promoted the production and quality of biochar, which decreased the bio-oil yield to 12.80, 12.97, and 9.50 wt% and increased the biochar yield to 78.87, 79.47, and 78.50 wt%, respectively. However, the usage of other catalysts like MgO, Fe₂O₃, MnO₂, CuCl₂, and NaCl enhanced the biogas yield and reduced the bio-oil yield. This was due to the poor microwave absorptivity of CaO, CuO, and MgCl₂, which hampered the rise of the temperature and led to the carbonization process. In contrast, other catalysts resulted in a high temperature easily and accelerated the decomposition of condensable gas into non-condensable gas. Vu Ly et al. (2022)

Table 8

Catalytic effect on the pyrolysis of food waste.

Biomass	Catalysts	Conditions	Reactor	Catalyst performance	Reference
Food waste	Pt/C	30–900 °C, 10 °C/min, N ₂ /CO ₂	TGA	- Enhances the production of non-condensable gas. - Accelerates the reduction of cyclic compounds.	(Mong et al., 2020a)
Food waste	Pt/AC	300–700 °C, N ₂	Fixed-bed reactor	- Increases the biogas yield and decreases the bio-oil yield. - Suppresses the formation of benzene derivatives and nitrile.	(Kim et al., 2020a)
Food waste	CaO, MgO, Fe ₂ O ₃ , MnO ₂ , CuCl ₂ , NaCl, MgCl ₂	300–600 W, N ₂	Microwave reactor	- MgO, Fe₂O₃, MnO₂, CuCl₂, and NaCl lower the bio-oil yield but raise the biogas yield. - MgO and MnO₂ reduce the biochar LHV and increase the pH values of lower layer bio-oil, while CuCl₂ and NaCl have the opposite effect. - CuCl₂ is the optimal catalyst based on the maximum energy ratio of production to consumption.	(Liu et al., 2014b)
Spent coffee waste	Dolomite, HZSM-5, Hematite, Magnetite	500 °C, 100 kg/h, N ₂	Fluidized-bed reactor	- The addition of catalyst ameliorates the bio-oil quality remarkably. - HZSM-5 reduces the oxygen content sharply and increases the HHV of bio-oil to 40.46 MJ/kg. - Dolomite and Hematite increase the bio-oil yield.	(Vu Ly et al., 2022)
Food waste/ <i>Chlorella vulgaris</i>	Permutit, CaO, CaCO ₃ , SiO ₂	900 °C, 20–50 °C/min, N ₂	TGA	- Permutit lowers the initial temperature, prolongs the main stage mass loss, and lessens the residues effectively. - CaO lessens the acids content and increases the hydrocarbons content. - CaO is the optimal catalyst for the pyrolysis of food waste.	(Chen et al., 2018)

employed Dolomite, HZSM-5, Hematite, and Magnetite to pyrolyzed spent coffee waste and detected that the accession of Dolomite and Hematite raised the bio-oil yield from 52.52 to 54.77 wt% and 54.04 wt%, respectively. While the presence of HZSM-5 and Magnetite cut down the bio-oil yield to 50.10 wt% and 51.64 wt%, respectively. The catalytic cracking of vapor induced by decarboxylation, dehydration, and decarbonylation reactions is the main cause of the decline in bio-oil yield. Moreover, Mong et al. (2020a) and Kim et al. (2020a) also confirmed that adding catalyst inhibited the formation of bio-oil but facilitated the generation of biogas with the help of Pt/C and Pt/AC, respectively.

Furthermore, the application of catalysts exerts a huge impact on the thermal degradation characteristics of food waste, permutit in particular, which has a promoting effect on reducing the initial temperature, increasing the main stage mass loss, and decreasing the residue mass (Chen et al., 2018). Additionally, the usage of catalysts affected the bio-oil composition largely. Vu Ly et al. (2022) found that HZSM-5 possessed the best deoxygenation capacity which decreased the oxygenated compounds from 15.47 to 12.55 wt% during the pyrolysis of spent coffee waste, the decarbonylation and dehydration reactions induced by HZSM-5 boosted the elimination of oxygen atoms in the form of moisture and CO in oxygenated compounds. The result was also verified by Yathavan and Agblevor (2013) and Liu et al. (2017). Fig. 7 exhibits the effect of catalysts on the composition of bio-oil originated from food waste pyrolysis and co-pyrolysis with microalgae.

Hydrocarbons like PAHs, benzene series, oxygenated compounds including saccharides, acids, alcohols, furans, and nitrogenous compounds are the primary constituents of bio-oil from the sole pyrolysis of food waste. Owing to the strong deacidification, deoxygenation, and admirable selectivity of hydrocarbons, CaO showed the best catalytic

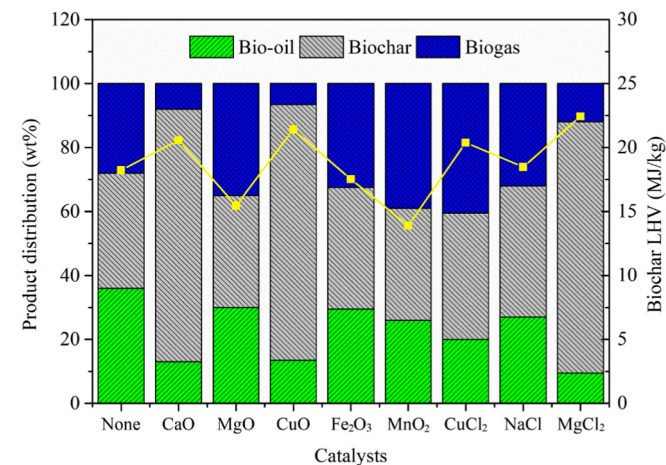
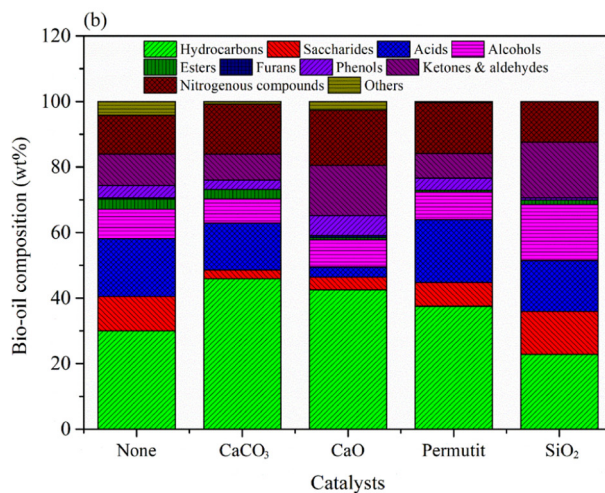
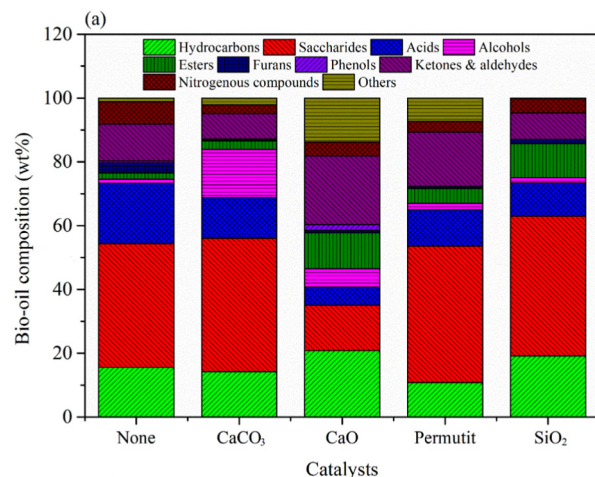


Fig. 6. The effect of catalysts on the product distribution and the biochar LHV (Liu et al., 2014b).

Fig. 7. The effect of catalysts on the composition of bio-oil from food waste (a) sole pyrolysis and (b) co-pyrolysis with microalgae (Chen et al., 2018).

performance on the improvement of hydrocarbons and the reduction of acids, which increased hydrocarbons content by 32.5% and 59.0%, decreased acids content by 70.15% and 82.08% during the pyrolysis and co-pyrolysis, respectively. The conclusion was consistent with other studies by Ding et al. (2018) and Wang et al. (2010). Additionally, CaCO_3 also presented a favorable catalytic effect on the bio-oil quality, especially for co-pyrolysis. Fig. 7 also demonstrated the superiority of co-pyrolysis, as the hydrocarbons content experienced a sharp rise and oxygenated compounds showed a remarkable decline after adding microalgae. Therefore, catalytic co-pyrolysis is the most promising technique to convert food waste into bioenergy.

There are some researches on catalytic pyrolysis of food waste available in the literature, however, other widely applied catalysts like bio-char, MCM-41, MnO_2 , Al_2O_3 , and Y zeolite have not been studied (Chan and Tanksale, 2014; Liu et al., 2014a). Furthermore, other advanced catalyst technologies, such as chemical liquid deposition and dual catalysts, can be used in the pyrolysis of food waste and further enhance their catalytic activity (Khoo et al., 2020).

5. Prospects and future challenges

In consideration of the increasing amount of food waste and the shortage of fossil fuels, pyrolysis is an effective method to reduce the amount of food waste and generate valuable products compared to conventional disposal methods of food waste (Chong et al., 2019; Lee et al., 2020b). Bio-oil originated from the pyrolysis of food waste is utilized to substitute for fossil fuels. Biochar is widely used as an adsorbent, catalyst, fertilizer, photocatalytic carrier, and anode materials (Goh et al., 2020; Zaker et al., 2019). Both bio-oil and bio-char are excellent products of energy recovery through thermochemical conversion and other processes.

Many researchers have dedicated efforts to study the economy and feasibility of food waste pyrolysis. Elkhailifa et al. (2019) compared several disposal methods, such as composting, landfilling, gasification, incineration, and pyrolysis to treat food waste and concluded that the pyrolysis process possessed significant economic and environmental superiorities owing to less infrastructure footprint, less time consuming, and the production of biofuel and biochar. Vu Ly et al. (2022) performed a techno-economic analysis on the catalytic pyrolysis of spent coffee waste for the production of high-quality bio-oil. Capital costs included pyrolysis reactor, combustor, cyclone, compressor and pump, storage and supplement equipment. Operating costs consisted of feedstock, silica sand for non-catalytic pyrolysis or catalyst for catalytic pyrolysis, energy, labor, and maintenance costs. After calculation and analysis, the production costs for the catalytic pyrolysis of spent coffee waste with silica, HZSM-5, dolomite, hematite, and magnetite were 0.0151, 0.0143, 0.0095, 0.0034, and 0.0102 USD/MJ, respectively, suggesting that the application of low-cost catalysts including dolomite and hematite exhibited better economic performance than fossil fuel-based crude oil (0.0098 USD/MJ). Thus, they concluded that bio-oil production through the catalytic fast pyrolysis of spent coffee waste was feasible and economical.

However, pretreatment techniques are essential to achieve a better pyrolysis result since the complex nature and high moisture content of food waste. High moisture content is not conducive to the generation of high-grade bio-oil (Khanh Nguyen et al., 2021; Kumari and Singh, 2018). Moreover, on the basis of surface chemistry, small particles size favors the improvement of reaction efficiency and bio-oil yield. Thus, drying and grinding are needed to be performed before pyrolysis. While pretreatment is an energy-intensive process, energy costs will be huge during the process (Sankaran et al., 2020; Soltanian et al., 2020). However, due to the dissimilarity of heating principles, the microwave-assisted heating method can skip some pretreatment steps like drying since water is able to be efficaciously removed in pyrolysis because of its polar nature (Liew et al., 2018; Mong et al., 2020b). Moreover, pyrolysis under vacuum conditions has better economic viability

owing to the lack of inert gas. In addition, the negative pressure is instrumental in the reduction of the boiling point, which benefits the formation of bio-oil (Liew et al., 2019; Wan Mahari et al., 2020). Lam et al. (2019) pyrolyzed waste cooking oil and plastics under N_2 environment or vacuum conditions and pointed out that the production cost in vacuum condition was 20.47% less than that under N_2 environment.

As an endothermic process, the energy content of final products is supposed to exceed that of feedstock because of the endothermic characteristics of pyrolysis. Meanwhile, synergistic effects between various materials in food waste are vital to achieving the exergy surplus (Tang et al., 2020). Nevertheless, the main impediment of synergistic effects is short of knowledge in a multi-component complex reaction network of food waste pyrolysis in the highly heterogeneous mixture environment. The mechanism of synergistic effects behind the food waste pyrolysis requires further exploration (Lee et al., 2021a; Lu et al., 2020).

Meanwhile, the enhancement of bio-oil quality can be realized through the addition of co-feedstocks and catalysts. The utilization of co-feedstocks with a high volatile matter content is instrumental in the production of bio-oil. The application of co-feedstock with high EHI is prone to induce a hydrogen transfer reaction, which is beneficial to the reduction of oxygenated compounds and the increment of hydrocarbons content (Ahmed et al., 2020; Gouws et al., 2021). Moreover, the presence of catalysts positively impacts product distribution and product properties via the promotion of dehydrogenation, denitrogenation, and aromatization reactions. Some studies have proven that the accession of catalysts poses a negative effect on the generation of coke and PAHs (Azizi et al., 2018; Liu et al., 2020b). As for the pyrolysis process, the usage of catalysts is good for decreasing activation energy and increasing reaction efficiency, resulting in the reduction of energy input (Zhang et al., 2016).

In addition, the deleterious effect of nitrogen-related pollution aroused by food waste on the environment cannot be ignored. Excess nitrogen in the atmosphere brings about a range of environmental issues, including ocean acidification, climate change, eutrophication of water bodies, ozone depletion, groundwater pollution, and toxic ground-level ozone pollution, which constituted a serious risk to the natural environment and human beings (Stevens, 2019).

Food waste possesses high nitrogen content owing to the existence of proteins, while the employment of pyrolysis is an effective method to control nitrogen pollution through the adjustment of operating parameters or the accession of catalysts. Ming et al. (2020) performed the thermal degradation experiment of food waste and found nitrogen mainly existed in the form of nitrile and amine because those two species possessed better thermal stability and were difficult to decompose at higher temperatures. Kim et al. (2020a) detected the addition of Pt/AC decreased the nitrile content largely during the pyrolysis of food waste. Patra et al. (2021) conducted the pyrolysis of food waste and observed that the increment of temperature reduced the nitrogen content in biochar. Deamination reaction induced by high temperature (600 °C) was the main reason. However, nitrogen content in bio-oil experienced an upward trend with the growth of temperature, upgrading technology like hydrodenitrogenation is required to be implemented to remove those heteroatoms. Similarly, Xu et al. (2020a) detected that the rise of temperature increased the content of nitrogenous species like quino-line, indole, and benzo[c]cinnoline to 3.20%, 6.15%, and 4.67% in bio-oil, but declined the nitrogen content in biochar. The secondary cracking caused by high temperature promotes the transfer of unstable nitrogen-containing compounds in biochar to bio-oil. Additionally, the high content of H_2 in carrier gas could hinder the generation of nitrogen-containing heterocycles in bio-oil. This was ascribed to the transformation from polycyclic nitrogen-containing heterocycles to monocyclic nitrogen-containing heterocycles was enhanced by H_2 , the ring-condensation or polymerization reaction between heteromonocycles were inhibited by high temperature and H^+ radicals. Moreover, a waste gas treating system can be built in the pyrolysis facility to remove those hazardous and toxic pollutants.

The promulgation and implementation of related laws, regulations, and policies further promote the resource utilization of food waste. China has issued notice on organizing food waste utilization and safe disposal pilots, selecting some pilot cities to carry out the project and formulating policies, regulations, and technologies in 2010. By 2015, approximately 100 pilot cities have participated in the project. Moreover, 10.9 billion CNY was expected to invest in establishing the food waste resource utilization facilities in the 12th Five-year Plan of China (Jin et al., 2021). In 2013, the Environmental Protection Agency and the United States Department of Agriculture launched the “US Food Waste Challenge”, built a platform for businesses and food service operations to decrease the environmental footprint of food waste by sustainable business practices, like purchases reduction, donations of additional food, and composting. Moreover, a law has been passed in France to forbid retailers to discard food and stipulates that all unsold but edible food be given to the charity for prompt distribution to the poor (Vogliano and Brown, 2016).

Furthermore, as shown in Fig. 8, many researchers have pointed out that although some large scale biomass energy production systems or centralized factory possesses more bargaining power in both the supply chain and sales chain due to the great demand for raw materials and the high and stable output of products. Nevertheless, the centralized factory is still confronted with many challenges, like huge capital input, complicated transportation and logistics, and high transportation and running fees. However, the superiority of small distributed plants can remedy these deficiencies. Firstly, the capital input and economic threshold of the small distributed plant are low and make it more attractive in those developing and underdeveloped countries, and the sharing of advanced technologies in developed countries will assist in the promotion of technical progress in underdeveloped countries. Resasco et al. (2018) reported that establishing those small plants would bring a better balance in the carbon, water, nitrogen, and phosphorus cycle. This will, in turn, ameliorate the local environment. Moreover, scattered factories can effectively shorten the transportation distance and reduce the transportation fee and environmental impact. Ruan et al. (2008) compared a centralized plant with small distributed plants and stated that small distributed plants held obvious advantages because of low capital costs, simplified transportation and logistics, and relatively high returns. In consideration of the dispersiveness in the distribution of food waste

resources, small distributed plants are more economical and viable. In real practice, the selection of plant facilities also needs to consider the actual local conditions, such as population, regional factors, dietary habits, relevant laws and regulations, the yield and characteristics of food waste, and environmental factors (Salleh et al., 2021).

6. Conclusions

This review concentrates on the conventional and advanced pyrolysis of food waste for the recovery of bioenergy. Pyrolysis is the optimal technique for dealing with food waste and producing valuable products. The pyrolysis mechanism of food waste has been analyzed and compared with other typical materials like plastics, tires, and lignocellulosic biomass. Bio-oil is the primary product during conventional pyrolysis under optimal conditions. The application of PVC, tire waste, microalgae, and lignocellulose biomass alters the distribution and properties of products of food waste pyrolysis. The usage of tire waste shows a favorable performance in the reduction of oxygenated compounds and the increment of hydrocarbons in bio-oil. The accession of microalgae also positively impacts the composition of bio-oil by lowering the acids level and raising the content of hydrocarbons and nitrogenous compounds. The addition of catalysts influences the pyrolysis process and product properties as well. The presence of CaO, CuO, and MgCl₂ increases the bio-oil yield and decreases the biochar yield effectively, while CaO and CaCO₃ exhibit an excellent promoting effect on the formation of hydrocarbons. Moreover, microwave-assisted catalytic co-pyrolysis of food waste under vacuum conditions presents great commercial and industrial potential. Small distributed plants are more economical and feasible in the treatment of food waste, especially in developing countries. In summary, pyrolysis is an excellent technique for achieving clean disposal and valorization of food waste.

CRediT authorship contribution statement

Guangcan Su: Conceptualization, Formal analysis, Writing - original draft. **Hwai Chyuan Ong:** Conceptualization, Supervision, Resources, Writing - reviewing & editing. **I. M. Rizwanul Fattah:** Validation, Writing - reviewing & editing. **Yong Sik Ok:** Formal analysis, Revision

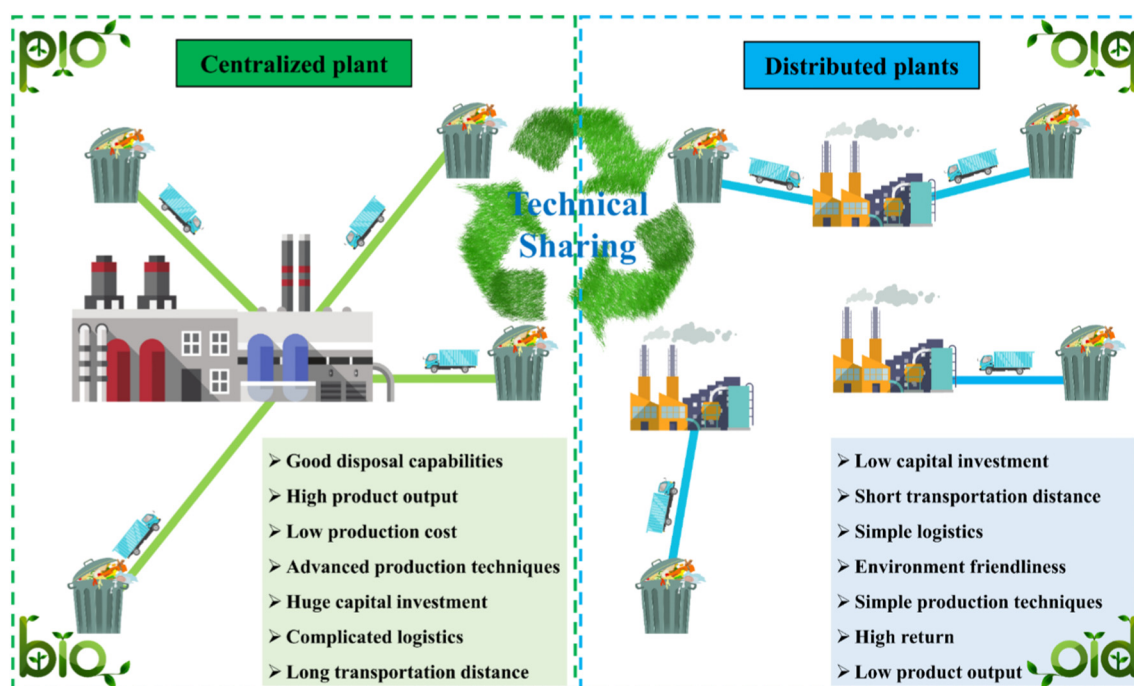


Fig. 8. Characteristics of centralized plants and small distributed plants (Resasco et al., 2018).

and formatting of paper. **Jer-Huan Jang**: Investigation, Writing - reviewing & editing. **Chin-Tsan Wang**: Visualization, Writing - reviewing & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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